

Perturbative Momentum Shell Renormalization Group

Luca Dall'Asta

DISAT
Politecnico di Torino, Italy

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Perturbation Theory: general approach

In the presence of (sufficiently small) non-quadratic terms, the expectation value of an operator $\mathcal{O}[\hat{\phi}]$ can be computed perturbatively,

$$\langle \mathcal{O} \rangle = \frac{\int D\hat{\phi} \mathcal{O}(\hat{\phi}) e^{-S[\hat{\phi}]}}{\int D\hat{\phi} e^{-S[\hat{\phi}]}} \quad (1)$$

$$= \frac{\int D\hat{\phi} \mathcal{O}(\hat{\phi}) e^{-S_0[\hat{\phi}] - S_I[\hat{\phi}]}}{\int D\hat{\phi} e^{-S_0[\hat{\phi}] - S_I[\hat{\phi}]}} \quad (2)$$

$$= \frac{Z_0^{-1} \int D\hat{\phi} \mathcal{O}(\hat{\phi}) e^{-S_0[\hat{\phi}]} \left\{ 1 - S_I + \frac{1}{2} S_I^2 - \dots \right\}}{Z_0^{-1} \int D\hat{\phi} e^{-S_0[\hat{\phi}]} \left\{ 1 - S_I + \frac{1}{2} S_I^2 - \dots \right\}} \quad (3)$$

$$= \frac{\langle \mathcal{O} \rangle_0 - \langle \mathcal{O} S_I \rangle_0 + \frac{1}{2} \langle \mathcal{O} S_I^2 \rangle_0 - \dots}{1 - \langle S_I \rangle_0 + \frac{1}{2} \langle S_I^2 \rangle_0 - \dots} \quad (4)$$

Perturbation Theory: general approach

$$= \left\{ \langle \mathcal{O} \rangle_0 - \langle \mathcal{O} S_I \rangle_0 + \frac{1}{2} \langle \mathcal{O} S_I^2 \rangle_0 - \dots \right\} \sum_{\ell \geq 0} \left[\langle S_I \rangle_0 - \frac{1}{2} \langle S_I^2 \rangle_0 - \dots \right]^\ell \quad (5)$$

$$= \left\{ \langle \mathcal{O} \rangle_0 - \langle \mathcal{O} S_I \rangle_0 + \frac{1}{2} \langle \mathcal{O} S_I^2 \rangle_0 - \dots \right\} \\ \times \left\{ 1 + \langle S_I \rangle_0 - \frac{1}{2} \langle S_I^2 \rangle_0 + \langle S_I \rangle_0^2 + \dots \right\} \quad (6)$$

$$= \langle \mathcal{O} \rangle_0 - [\langle \mathcal{O} S_I \rangle_0 - \langle \mathcal{O} \rangle_0 \langle S_I \rangle_0] \\ + \frac{1}{2} [\langle \mathcal{O} S_I^2 \rangle_0 - 2 \langle \mathcal{O} S_I \rangle_0 \langle S_I \rangle_0 + 2 \langle \mathcal{O} \rangle_0 \langle S_I \rangle_0^2 - \langle \mathcal{O} \rangle_0 \langle S_I^2 \rangle_0] + \dots \quad (7)$$

These terms can be identified with *connected averages* (cumulants) over the gaussian measure $e^{-S_0[\hat{\phi}]}$,

$$\Rightarrow \quad \langle \mathcal{O} \rangle = \sum_{\ell \geq 0} \frac{(-1)^\ell}{\ell!} \langle \mathcal{O} S_I^\ell \rangle_{0,c} \quad (8)$$

Perturbation Theory: scalar ϕ^4 field theory

Consider the Landau-Ginzburg effective Hamiltonian (or action):

$$\beta\mathcal{H}_{\text{eff}}[\phi] = S[\phi] = \int d^d x \left[\frac{c}{2} (\nabla\phi)^2 + \frac{rt}{2} \phi(\vec{x})^2 + \frac{u}{4} \phi(\vec{x})^4 \right], \quad (9)$$

The Fourier representation of the action is

$$S[\hat{\phi}] = \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}(\vec{k})|^2 + \frac{u}{4} \int d^d x \left[\int \frac{d\vec{k}}{(2\pi)^d} \hat{\phi}(\vec{k}) e^{i\vec{k}\cdot\vec{x}} \right]^4 \quad (10)$$

$$\begin{aligned} &= \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}(\vec{k})|^2 + \frac{u}{4} \int d^d x \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} e^{i\vec{x}\cdot\sum_i \vec{k}_i} \prod_{i=1}^4 \hat{\phi}(\vec{k}_i) \\ &= \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}(\vec{k})|^2 \\ &\quad + \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_{i=1}^4 \vec{k}_i\right) \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4), \end{aligned} \quad (11)$$

Corrections to the two-point correlation function

We identify the fourth-order term as the interacting part of the action,

$$S_I[\hat{\phi}] = \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \quad (12)$$

Setting $\mathcal{O}(\hat{\phi}) = \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}')$, its expected value is given by

$$\langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}') \rangle = \langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}') \rangle_0 - \left[\langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}') S_I \rangle_0 - \langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}') \rangle_0 \langle S_I \rangle_0 \right] + O(u^2) \quad (13)$$

$$\begin{aligned} &= \langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}') \rangle_0 \\ &\quad - \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}')\hat{\phi}(\vec{k}_1)\hat{\phi}(\vec{k}_2)\hat{\phi}(\vec{k}_3)\hat{\phi}(\vec{k}_4) \rangle_0 \\ &\quad + \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \langle \hat{\phi}(\vec{k})\hat{\phi}(\vec{k}') \rangle_0 \langle \hat{\phi}(\vec{k}_1)\hat{\phi}(\vec{k}_2)\hat{\phi}(\vec{k}_3)\hat{\phi}(\vec{k}_4) \rangle_0 \\ &\quad + O(u^2) \end{aligned} \quad (14)$$

Corrections to the two-point correlation function

Using Wick's theorem,

$$\begin{aligned} & \langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \rangle_0 \\ &= \langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \rangle_0 \langle \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \rangle_0 \\ & \quad + [\langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}_1) \rangle_0 \langle \hat{\phi}(\vec{k}') \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \rangle_0 + 3 \text{ perm.}] \\ &= \langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \rangle_0 \langle \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \rangle_0 \\ & \quad + [\langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}_1) \rangle_0 \langle \hat{\phi}(\vec{k}') \hat{\phi}(\vec{k}_2) \rangle_0 \langle \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \rangle_0 + 11 \text{ perm.}] \\ &= \langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \rangle_0 \langle \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4) \rangle_0 \\ & \quad + [(2\pi)^d \delta(\vec{k} + \vec{k}_1) \hat{G}_0(\vec{k}_1) (2\pi)^d \delta(\vec{k}_2 + \vec{k}') \hat{G}_0(\vec{k}_2) (2\pi)^d \delta(\vec{k}_3 + \vec{k}_4) \hat{G}_0(\vec{k}_3) \\ & \quad + 11 \text{ perm.}] \end{aligned} \tag{15}$$

Corrections to the two-point correlation function

Using

$$\langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \rangle = (2\pi)^d \delta(\vec{k} + \vec{k}') \hat{G}(\vec{k}), \quad (16)$$

$$\langle \hat{\phi}(\vec{k}) \hat{\phi}(\vec{k}') \rangle_0 = (2\pi)^d \delta(\vec{k} + \vec{k}') \hat{G}_0(\vec{k}) = \frac{(2\pi)^d \delta(\vec{k} + \vec{k}')}{rt + ck^2} \quad (17)$$

we obtain

$$\begin{aligned} \hat{G}(\vec{k}) &= \hat{G}_0(k) - \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} \delta\left(\sum_{i=1}^4 \vec{k}_i\right) (2\pi)^{3d} \\ &\quad \times \delta(\vec{k} + \vec{k}_1) \hat{G}_0(k_1) \delta(\vec{k}_2 - \vec{k}) \hat{G}_0(k_2) \delta(\vec{k}_3 + \vec{k}_4) \hat{G}_0(k_3) \\ &\quad + 11 \text{ perm.} + O(u^2) \end{aligned} \quad (18)$$

$$= \hat{G}_0(k) - \frac{12u}{4} [\hat{G}_0(k)]^2 \int^\Lambda \frac{d\vec{q}}{(2\pi)^d} \hat{G}_0(q) + O(u^2) \quad (19)$$

$$= \frac{1}{rt + ck^2} \left\{ 1 - \frac{3u}{rt + ck^2} \int^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \right\} + O(u^2) \quad (20)$$

Corrections to the critical temperature

The susceptibility is given by $\chi = \int d\vec{r} G(\vec{r}) = \hat{G}(\vec{k} = 0)$, i.e.

$$\chi = \frac{1}{rt} \left\{ 1 - \frac{3u}{rt} \int^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} + O(u^2) \right\} \quad (21)$$

Consider χ^{-1} , then

$$\Rightarrow \chi^{-1} = rt \left\{ 1 - \frac{3u}{rt} \int^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} + O(u^2) \right\}^{-1} \quad (22)$$

$$= rt \left\{ 1 + \frac{3u}{rt} \int^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} + O(u^2) \right\} \quad (23)$$

$$= rt + 3u \int^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} + O(u^2) \quad (24)$$

Corrections to the critical temperature

At the MF critical point $t = 0$, we find $\chi^{-1}(t = 0) \neq 0$ because the critical temperature is shifted. Instead,

$$0 = \chi^{-1}(t_c) = rt_c + 3u \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt_c + cq^2} + O(u^2) \quad (25)$$

$$\Rightarrow t_c = -\frac{3u}{r} \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt_c + cq^2} + O(u^2) \quad (26)$$

$$\approx -\frac{3u}{rc} K_d \frac{\Lambda^{d-2}}{d-2} < 0, \quad d > 2 \quad (27)$$

where $K_d = S_d/(2\pi)^d$ and $S_d = \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})}$ is the surface of the d -dim sphere coming from the angular integration.

Divergence of the susceptibility

We can compute how the susceptibility diverges at criticality,

$$\chi^{-1}(t) - \chi^{-1}(t_c) = rt + 3u \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} - rt_c - 3u \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt_c + cq^2} \quad (28)$$

$$= r(t - t_c) + 3u \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \left[\frac{1}{rt + cq^2} - \frac{1}{rt_c + cq^2} \right] \quad (29)$$

$$= r(t - t_c) - 3u \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{r(t - t_c)}{(rt_c + cq^2)(r(t - t_c) + rt_c + cq^2)} \quad (30)$$

$$\approx r(t - t_c) \left\{ 1 - \frac{3u}{c^2} \int^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{q^2 (r(t - t_c)/c + q^2)} \right\} \quad (31)$$

$$\approx r(t - t_c) \left\{ 1 - \frac{3uK_d}{c^2} \left(\frac{r(t - t_c)}{c} \right)^{\frac{d}{2}-2} \int_0^{\Lambda} \sqrt{\frac{c}{r(t-t_c)}} \frac{dx x^{d-3}}{(1+x^2)} \right\} \quad (32)$$

The correction term diverges at criticality for $d < 4$, meaning that in this regime the perturbation series is inapplicable.

Validity of perturbative approach

The parameter u is not a “good” expansion parameter because $[u] = \Lambda^{-d}$. In fact, we also have $[c] = \Lambda^{-2}$, and $[r(t - t_c)/c] = \Lambda^2$, therefore a possible non-dimensional *expansion parameter* is

$$\check{u} = \frac{u}{c^2} \left(\frac{r(t - t_c)}{c} \right)^{d/2-2} \quad (33)$$

- for $d > 4$, $\check{u} \rightarrow 0$ as $t \rightarrow t_c$, which justifies accuracy of MF theory above upper critical dimension;

REMARK: The point $u = 0$ is not regular, the perturbation theory can at most be defined as an *asymptotic series*.

- for $d < 4$, $\check{u} \rightarrow \infty$ as $t \rightarrow t_c$, and perturbation theory appears ill-defined.

REMARK: However, in the series expansion $e^{-x} = \sum_{n \geq 0} \frac{(-x)^n}{n!}$, each term diverges for $x \rightarrow \infty$, but the series is finite.

Perturbative momentum-shell approach

In order to avoid the onset of IR divergencies, at criticality, in the corrections terms, we can employ Wilson's momentum shell RG approach, which only evaluates integrals for $\Lambda/\ell < |\vec{k}| < \Lambda$.

Consider again the general action of the ϕ^4 field theory:

$$S[\hat{\phi}] = \int^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}(\vec{k})|^2 + \frac{u}{4} \int d^d x \left[\int \frac{d\vec{k}}{(2\pi)^d} \hat{\phi}(\vec{k}) e^{i\vec{k} \cdot \vec{x}} \right]^4 \quad (34)$$

$$\begin{aligned} &= \int^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}(\vec{k})|^2 + \frac{u}{4} \int d^d x \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} e^{i\vec{x} \cdot \sum_i \vec{k}_i} \prod_{i=1}^4 \hat{\phi}(\vec{k}_i) \\ &= \int^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}(\vec{k})|^2 \\ &\quad + \frac{u}{4} \int^{\Lambda} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_i \vec{k}_i \right) \hat{\phi}(\vec{k}_1) \hat{\phi}(\vec{k}_2) \hat{\phi}(\vec{k}_3) \hat{\phi}(\vec{k}_4), \end{aligned} \quad (35)$$

Perturbative momentum-shell approach

Introducing field decomposition in fast and slow modes, the coarse-grained measure contains a non-trivial interaction term

$$e^{-S'[\hat{\phi}_<]} = e^{-S_0[\hat{\phi}_<]} \int D\hat{\phi}_> e^{-S_0[\hat{\phi}_>] - S_I[\hat{\phi}_>, \hat{\phi}_<]} \quad (36)$$

with

$$S_0[\hat{\phi}_<] = \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} |\hat{\phi}_<(\vec{k})|^2 \quad (37)$$

$$S_I[\hat{\phi}_>, \hat{\phi}_<] = \frac{u}{4} \int^{\Lambda} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_i \vec{k}_i\right) \left[\hat{\phi}_<(\vec{k}_1) + \hat{\phi}_>(\vec{k}_1)\right] \quad (38) \\ \times \left[\hat{\phi}_<(\vec{k}_2) + \hat{\phi}_>(\vec{k}_2)\right] \left[\hat{\phi}_<(\vec{k}_3) + \hat{\phi}_>(\vec{k}_3)\right] \left[\hat{\phi}_<(\vec{k}_4) + \hat{\phi}_>(\vec{k}_4)\right]$$

Perturbative momentum-shell approach

Introduce the factor

$$Z_0^> = \int D\hat{\phi}_> e^{-S_0[\hat{\phi}_>]} \quad (39)$$

Rewrite the coarse-grained measure as follows

$$\begin{aligned} e^{-S'[\hat{\phi}_<]} &= e^{-S_0[\hat{\phi}_<]} Z_0^> \\ &\quad \times \frac{1}{Z_0^>} \int D\hat{\phi}_> e^{-S_0[\hat{\phi}_>] - S_I[\hat{\phi}_>, \hat{\phi}_<]} \end{aligned} \quad (40)$$

$$= Z_0^> e^{-S_0[\hat{\phi}_<]} \frac{\int D\hat{\phi}_> e^{-S_0[\hat{\phi}_>]} e^{-S_I[\hat{\phi}_>, \hat{\phi}_<]}}{\int D\hat{\phi}_> e^{-S_0[\hat{\phi}_>]}} \quad (41)$$

$$= Z_0^> e^{-S_0[\hat{\phi}_<]} \left\langle e^{-S_I[\hat{\phi}_>, \hat{\phi}_<]} \right\rangle_{0,>} \quad (42)$$

which implies

$$S'[\hat{\phi}_<] = S_0[\hat{\phi}_<] - \log \left\langle e^{-S_I[\hat{\phi}_>, \hat{\phi}_<]} \right\rangle_{0,>} \quad (43)$$

Perturbative momentum-shell approach

Notice that for small X , we have

$$\log \langle e^X \rangle = \log \left\langle 1 + X + \frac{X^2}{2} + \dots \right\rangle \quad (44)$$

$$= \langle X \rangle + \frac{1}{2} \langle X^2 \rangle - \frac{1}{2} \langle X \rangle^2 + O(X^3) \quad (45)$$

$$= \langle X \rangle + \frac{1}{2} \left(\langle X^2 \rangle - \langle X \rangle^2 \right) + O(X^3), \quad (46)$$

so assuming S_I (because of u) to be small,

$$\log \langle e^{-S_I[\hat{\phi}_{>}, \hat{\phi}_{<}]} \rangle_{0,>} = -\langle S_I \rangle_{0,>} + \frac{1}{2} \left[\langle S_I^2 \rangle_{0,>} - \langle S_I \rangle_{0,>}^2 \right] + \dots \quad (47)$$

$$= \sum_{\ell \geq 1} \frac{(-1)^\ell}{\ell!} M_\ell(S_I), \quad (48)$$

where $M_\ell(S_I)$ is the ℓ -th cumulant of S_I .

First-order contributions to the coarse-grained action

The first-order contribution is given by

$$\begin{aligned} \langle S_I \rangle_{0,>} &= \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_i \vec{k}_i \right) \langle (\hat{\phi}_<(\vec{k}_1) + \hat{\phi}_>(\vec{k}_1)) \\ &\quad \times (\hat{\phi}_<(\vec{k}_2) + \hat{\phi}_>(\vec{k}_2)) (\hat{\phi}_<(\vec{k}_3) + \hat{\phi}_>(\vec{k}_3)) (\hat{\phi}_<(\vec{k}_4) + \hat{\phi}_>(\vec{k}_4)) \rangle_{0,>}. \quad (49) \end{aligned}$$

There are $2^4 = 16$ terms, combining fast and slow fields:

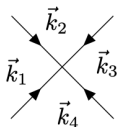
- 1 term with 4 slow fields,
- 4 terms with 3 slow fields and 1 fast field,
- 6 terms with 2 slow fields and 2 fast fields,
- 4 terms with 1 slow field and 3 fast fields,
- 1 term with 4 fast fields.

First-order contributions to the coarse-grained action

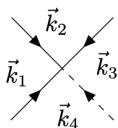
Slow and fast fields can be represented graphically as follows

$$\hat{\phi}_{<}(\vec{k}) = \longrightarrow, \quad \hat{\phi}_{>}(\vec{k}) = \dashrightarrow$$

Graphical representation of the 16 “vertices”:

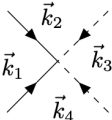
$$\hat{\phi}_{<}(\vec{k}_1)\hat{\phi}_{<}(\vec{k}_2)\hat{\phi}_{<}(\vec{k}_3)\hat{\phi}_{<}(\vec{k}_4) =$$


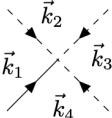
The diagram shows a central vertex where four solid lines meet. The lines are labeled with momentum vectors: $\vec{k}_1$ (bottom-left, incoming), $\vec{k}_2$ (top-left, incoming), $\vec{k}_3$ (top-right, incoming), and $\vec{k}_4$ (bottom-right, incoming). All lines have arrows pointing towards the central vertex.

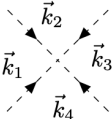
$$\hat{\phi}_{<}(\vec{k}_1)\hat{\phi}_{<}(\vec{k}_2)\hat{\phi}_{<}(\vec{k}_3)\hat{\phi}_{>}(\vec{k}_4) + 3 \text{ perm.} =$$


The diagram shows a central vertex where four lines meet. Three lines are solid and labeled $\vec{k}_1$ (bottom-left, incoming), $\vec{k}_2$ (top-left, incoming), and $\vec{k}_3$ (top-right, incoming). The fourth line, labeled $\vec{k}_4$ (bottom-right, outgoing), is dashed. All solid lines have arrows pointing towards the vertex, while the dashed line has an arrow pointing away from the vertex.

First-order contributions to the coarse-grained action

$$\hat{\phi}_{<}(\vec{k}_1)\hat{\phi}_{<}(\vec{k}_2)\hat{\phi}_{>}(\vec{k}_3)\hat{\phi}_{>}(\vec{k}_4) + 5 \text{ perm.} =$$


$$\hat{\phi}_{<}(\vec{k}_1)\hat{\phi}_{>}(\vec{k}_2)\hat{\phi}_{>}(\vec{k}_3)\hat{\phi}_{>}(\vec{k}_4) + 3 \text{ perm.} =$$


$$\hat{\phi}_{>}(\vec{k}_1)\hat{\phi}_{>}(\vec{k}_2)\hat{\phi}_{>}(\vec{k}_3)\hat{\phi}_{>}(\vec{k}_4) =$$


Each term is averaged over the gaussian measure $e^{-S_0[\hat{\phi}_{>}]}$ for fast modes.

First-order contributions to the coarse-grained action

The first contribution completes the expression of the action $S[\hat{\phi}_<]$, i.e.

$$\begin{aligned} S[\hat{\phi}_<] &= S_0[\hat{\phi}_<] + \frac{u}{4} \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_i \vec{k}_i\right) \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_<(\vec{k}_4) \\ &= S_0[\hat{\phi}_<] + \frac{u}{4} \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_<(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \end{aligned} \quad (50)$$

The second contribution is zero, indeed

$$\left\langle \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_>(\vec{k}_4) \right\rangle_{0,>} = \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \left\langle \hat{\phi}_>(\vec{k}_4) \right\rangle_{0,>} \quad (51)$$

$$= \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \times 0 \quad (52)$$

$$= 0 \quad (53)$$

First-order contributions to the coarse-grained action

Similarly, the fourth contribution is zero because

$$\langle \hat{\phi}_{<}(\vec{k}_1) \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} = \hat{\phi}_{<}(\vec{k}_1) \langle \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \quad (54)$$

$$\stackrel{\text{Wick th.}}{=} \hat{\phi}_{<}(\vec{k}_1) \left\{ \langle \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \rangle_{0,>} \langle \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} + \langle \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \langle \hat{\phi}_{>}(\vec{k}_3) \rangle_{0,>} + \langle \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \langle \hat{\phi}_{>}(\vec{k}_2) \rangle_{0,>} \right\} \quad (55)$$

$$= 0 \quad (56)$$

The fifth contribution is given by

$$\begin{aligned} \langle \hat{\phi}_{>}(\vec{k}_1) \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} &= \langle \hat{\phi}_{>}(\vec{k}_1) \hat{\phi}_{>}(\vec{k}_2) \rangle_{0,>} \langle \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \\ &+ \langle \hat{\phi}_{>}(\vec{k}_1) \hat{\phi}_{>}(\vec{k}_3) \rangle_{0,>} \langle \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \\ &+ \langle \hat{\phi}_{>}(\vec{k}_1) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \langle \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \rangle_{0,>} \end{aligned} \quad (57)$$

First-order contributions to the coarse-grained action

Using $\langle \hat{\phi}_{>}(\vec{k}) \hat{\phi}_{>}(\vec{k}') \rangle_{0,>} = (2\pi)^d \delta(\vec{k} + \vec{k}') \hat{G}_{0,>}(k)$ we find

$$\begin{aligned} & \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_i \vec{k}_i\right) \langle \hat{\phi}_{>}(\vec{k}_1) \hat{\phi}_{>}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \\ &= \frac{u}{4} \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^d} \delta\left(\sum_i \vec{k}_i\right) \delta(\vec{k}_1 + \vec{k}_2) \hat{G}_{0,>}(\vec{k}_1) \delta(\vec{k}_3 + \vec{k}_4) \hat{G}_{0,>}(\vec{k}_3) + 2 \text{ perm.} \\ &= \frac{3u}{4} (2\pi)^d \delta(0) \left[\int_{\Lambda/\ell}^\Lambda \frac{d\vec{k}}{(2\pi)^d} \hat{G}_{0,>}(\vec{k}) \right]^2 \propto \frac{3u}{4} \left[\int_{\Lambda/\ell}^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{1}{rt+ck^2} \right]^2 V \end{aligned}$$

which has the diagrammatical representation,

$$\left[\int_{\Lambda/\ell}^\Lambda \frac{d\vec{k}}{(2\pi)^d} \hat{G}_{0,>}(\vec{k}) \right]^2 =$$


It gives an additive contribution to the action which will be neglected.

First-order contributions to the coarse-grained action

The third term instead is very important, up to 6 permutations,

$$\langle \hat{\phi}_{<}(\vec{k}_1) \hat{\phi}_{<}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} = \hat{\phi}_{<}(\vec{k}_1) \hat{\phi}_{<}(\vec{k}_2) \langle \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} \quad (58)$$

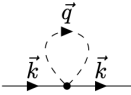
$$= \hat{\phi}_{<}(\vec{k}_1) \hat{\phi}_{<}(\vec{k}_2) (2\pi)^d \delta(\vec{k}_3 + \vec{k}_4) \hat{G}_{0,>}(\vec{k}_3) \quad (59)$$

Then the final contribution

$$\begin{aligned} & \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_i \vec{k}_i\right) \left[\langle \hat{\phi}_{<}(\vec{k}_1) \hat{\phi}_{<}(\vec{k}_2) \hat{\phi}_{>}(\vec{k}_3) \hat{\phi}_{>}(\vec{k}_4) \rangle_{0,>} + 5 \text{ perm.} \right] \\ &= \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \left[\hat{\phi}_{<}(\vec{k}_1) \hat{\phi}_{<}(\vec{k}_2) (2\pi)^d \delta(\vec{k}_1 + \vec{k}_2) \hat{G}_{0,>}(\vec{k}_3) + 5 \text{ perm.} \right] \\ &= 6 \int_{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \hat{\phi}_{<}(\vec{k}) \hat{\phi}_{<}(-\vec{k}) \int_{\Lambda/\ell} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt+cq^2} \end{aligned} \quad (60)$$

First-order contributions to the coarse-grained action

The diagrammatical representation is

$$\int_0^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \hat{\phi}_<(\vec{k}) \hat{\phi}_<(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} = \text{diagram}$$
A Feynman diagram representing a loop correction. It consists of a central vertex from which two solid lines extend horizontally to the left and right, each labeled with a vector $\vec{k}$ and an arrow pointing away from the vertex. A dashed line forms a loop above the vertex, with an arrow pointing clockwise and labeled with a vector $\vec{q}$.

This contribution provides a perturbative correction to the quadratic term in the action, which can be interpreted as a *renormalization of the parameter t* , i.e.

$$rt_1 = rt + 12 \frac{u}{4} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2}. \quad (61)$$

Second-order contributions to the coarse-grained action

Up to $O(u^2)$, we have

$$S' [\hat{\phi}_<] = S_0 [\hat{\phi}_<] + \langle S_I \rangle_{0,>} - \frac{1}{2} \left[\langle S_I^2 \rangle_{0,>} - \langle S_I \rangle_{0,>}^2 \right] + O(u^3) \quad (62)$$

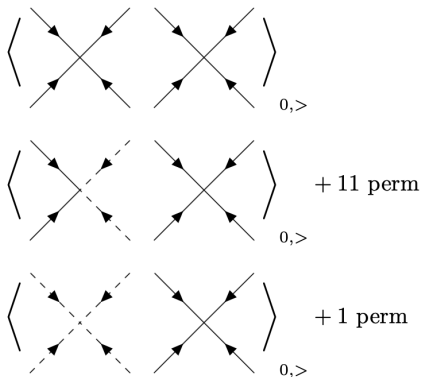
in which

$$\begin{aligned} \langle S_I^2 \rangle_{0,>} &= \left(\frac{u}{4} \right)^2 \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \int^\Lambda \frac{d\vec{k}_5 d\vec{k}_6 d\vec{k}_7 d\vec{k}_8}{(2\pi)^{4d}} (2\pi)^d \delta \left(\sum_{i=5}^8 \vec{k}_i \right) \\ &\quad \times \left\langle \left(\hat{\phi}_<(\vec{k}_1) + \hat{\phi}_>(\vec{k}_1) \right) \left(\hat{\phi}_<(\vec{k}_2) + \hat{\phi}_>(\vec{k}_2) \right) \left(\hat{\phi}_<(\vec{k}_3) + \hat{\phi}_>(\vec{k}_3) \right) \left(\hat{\phi}_<(\vec{k}_4) + \hat{\phi}_>(\vec{k}_4) \right) \right. \\ &\quad \times \left. \left(\hat{\phi}_<(\vec{k}_5) + \hat{\phi}_>(\vec{k}_5) \right) \left(\hat{\phi}_<(\vec{k}_6) + \hat{\phi}_>(\vec{k}_6) \right) \left(\hat{\phi}_<(\vec{k}_7) + \hat{\phi}_>(\vec{k}_7) \right) \left(\hat{\phi}_<(\vec{k}_8) + \hat{\phi}_>(\vec{k}_8) \right) \right\rangle_{0,>} \end{aligned} \quad (63)$$

The term $\langle S_I^2 \rangle$ generates $2^8 = 256$ contributions, among them $2^7 = 128$ non-vanishing terms containing an even number of fast modes.

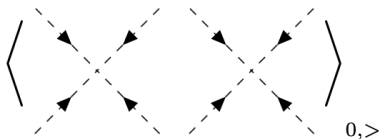
Second-order contributions to the coarse-grained action

It is useful to provide a graphical representation of these contributions. The following ones generate disconnected diagrams canceled by equal terms from $\langle S_I \rangle^2$.



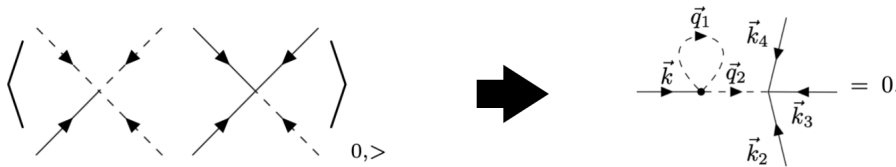
Second-order contributions to the coarse-grained action

The contributions from



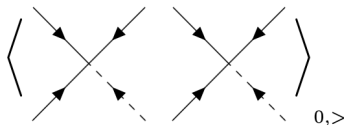
generate a constant additive term to the action of slow modes.

There are $2 \times 4 \times 4 = 32$ terms, each one generating 3 vanishing diagrams,



Second-order contributions to the coarse-grained action

Then we have $4 \times 4 = 16$ terms of type



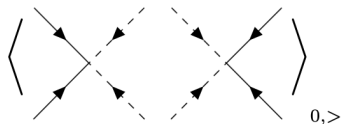
each one generating a diagram of type

$$\begin{aligned}
 & \begin{array}{c} \vec{k}_1 \\ \swarrow \\ \vec{k}_2 \rightarrow \text{---} \vec{q} \text{---} \\ \nwarrow \\ \vec{k}_3 \end{array} \begin{array}{c} \vec{k}_6 \\ \swarrow \\ \vec{k}_5 \leftarrow \text{---} \\ \nwarrow \\ \vec{k}_4 \end{array} = \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \delta(\vec{q} - \vec{k}_1 - \vec{k}_2 - \vec{k}_3) \hat{\phi}_<(\vec{k}_4) \hat{\phi}_<(\vec{k}_5) \hat{\phi}_<(\vec{k}_6) \\
 & \times \hat{G}_{0,>}(\vec{q}) \delta(\vec{q} + \vec{k}_4 + \vec{k}_5 + \vec{k}_6).
 \end{aligned}$$

a 6-leg vertex contribution to $S'[\hat{\phi}_<]$ can be neglected: (1) it is a very small contribution, (2) it is irrelevant under RG.

Second-order contributions to the coarse-grained action

Then we have $6 \times 6 = 36$ terms from contributions of type

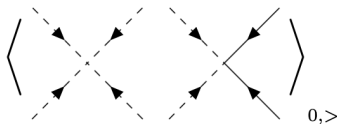


which generate a total of $36 \times 2 = 72$ diagrams as the following one,

$$\begin{aligned}
 & \begin{array}{c} \vec{k}_2 \quad \vec{k}_1 + \vec{k}_2 - \vec{q} \quad \vec{k}_3 \\ \swarrow \quad \searrow \quad \nearrow \\ \vec{q} \\ \nwarrow \quad \nearrow \quad \searrow \\ \vec{k}_1 \quad -\sum_{i=1}^3 \vec{k}_i \end{array} = \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \hat{G}_{0,>}(\vec{q}) \hat{G}_{0,>}(\vec{k}_1 + \vec{k}_2 - \vec{q}) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_<(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \\
 & \hspace{15em} (6.41a) \\
 & = \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_<(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \underbrace{\int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2) \left(rt + c \left(\vec{k}_1 + \vec{k}_2 - \vec{q} \right)^2 \right)}}_{=B(\vec{K};c,rt)}
 \end{aligned}$$

Second-order contributions to the coarse-grained action

Then we have $2 \times 6 = 12$ contributions



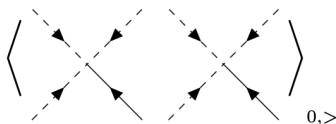
which generate $12 \times 4 \times 3 = 144$ diagrams of the type

$$= \hat{\phi}_{<}(\vec{k}) \hat{\phi}_{<}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \hat{G}_{0,>}(\vec{q}_1)^2 \hat{G}_{0,>}(\vec{q}_2),$$

This is a shift in the parameter t at $O(u^2)$.

Second-order contributions to the coarse-grained action

Then we have $4 \times 4 = 16$ contributions

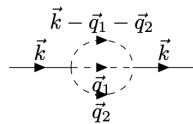


which generate: (a) $16 \times 3 \times 3 = 144$ vanishing diagrams of type

$$= \hat{\phi}_{<}(\vec{k}) \hat{\phi}_{<}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \hat{G}_{0,>}(\vec{q}_1) \hat{G}_{0,>}(\vec{k}) \hat{G}_{0,>}(\vec{q}_2) = 0$$

Second-order contributions to the coarse-grained action

and (b) $16 \times 3 \times 2 = 96$ diagrams of type



$$= \hat{\phi}_{<}(\vec{k}) \hat{\phi}_{<}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \hat{G}_{0,>}(\vec{q}_1) \hat{G}_{0,>}(\vec{q}_2) \hat{G}_{0,>}(\vec{k} - \vec{q}_1 - \vec{q}_2),$$

$$= \underbrace{\left| \hat{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{(rt + cq_1^2)(rt + cq_2^2) \left(rt + c \left(\vec{k} - \vec{q}_1 - \vec{q}_2 \right)^2 \right)}}_{=A(\vec{k};c,rt)},$$

Renormalized action up to second order

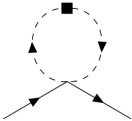
Considering all non-trivial contributions, we have

$$\begin{aligned}
 S'[\hat{\phi}_<] &= \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \left| \hat{\phi}_<(\vec{k}) \right|^2 \left[\frac{rt + ck^2}{2} + 6\frac{u}{4} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \right. \\
 &\quad \left. - 72 \left(\frac{u}{4} \right)^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{1}{(rt + cq_1^2)^2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{rt + cq_2^2} - \frac{96}{2} \left(\frac{u}{4} \right)^2 A(\vec{k}; c, rt) \right] \\
 &\quad + \frac{u}{4} \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_<(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \left[1 - 36 \frac{u}{4} B(\vec{k}_1 + \vec{k}_2; c, rt) \right] \quad (64) \\
 &= \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \left| \hat{\phi}_<(\vec{k}) \right|^2 \left[\frac{rt_1 + c_1 k^2}{2} + O(k^4) \right] \\
 &\quad + \frac{1}{4} \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} u_1 (\vec{k}_1 + \vec{k}_2) \hat{\phi}_<(\vec{k}_1) \hat{\phi}_<(\vec{k}_2) \hat{\phi}_<(\vec{k}_3) \hat{\phi}_<(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \quad (65)
 \end{aligned}$$

REMARK: Coarse-grained parameters depend on wavevectors (higher-order terms appear).

Renormalized action up to second order

Corrections to t are simple and come from



$$\begin{aligned}
 &= \left| \hat{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{1}{rt_1 + cq_1^2}, \\
 &= \left| \hat{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{1}{rt + 12\frac{u}{4} \int \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{rt + cq_2^2} + cq_1^2} \\
 &= \left| \hat{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{1}{rt + cq_1^2} - 12\frac{u}{4} \left| \hat{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{(rt + cq_1^2)^2 (rt + cq_2^2)}.
 \end{aligned}$$

and from

$$A(0; c, rt) = \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{(rt + cq_1^2) (rt + cq_2^2) \left(rt + c(\vec{q}_1 + \vec{q}_2)^2 \right)}. \quad (66)$$

Renormalized action up to second order

Corrections to c at $O(u^2)$ come from the ugly term

$$\left. \frac{d}{dk^2} A(\vec{k}; c, rt) \right|_{k^2=0} = \left. \frac{d}{dk^2} [A(\vec{k}; c, rt) - A(0; c, rt) + A(0; c, rt)] \right|_{k^2=0} \quad (67)$$

$$= \left. \frac{d}{dk^2} [A(\vec{k}; c, rt) - A(0; c, rt)] \right|_{k^2=0} \quad (68)$$

$$= \frac{d}{dk^2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{(rt + cq_1^2)(rt + cq_2^2)} \\ \times \left[\frac{1}{rt + c(\vec{k} - \vec{q}_1 - \vec{q}_2)^2} - \frac{1}{rt + c(\vec{q}_1 + \vec{q}_2)^2} \right] \Big|_{k^2=0} \quad (69)$$

For the moment we neglect it, but we will come back later.

Renormalized action up to second order

Corrections to u at $O(u^2)$ can be approximated as follows:

$$B(\vec{K}; c, rt) = \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2) \left(rt + c (\vec{K} - \vec{q})^2 \right)} \quad (70)$$

$$= \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2) \left(rt + c (K^2 + q^2 - 2\vec{K} \cdot \vec{q}) \right)} \quad (71)$$

$$= \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2)^2 \left(1 + \frac{cK^2 - 2c\vec{K} \cdot \vec{q}}{rt + cq^2} \right)} \quad (72)$$

$$\approx \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2)^2} \left(1 - \frac{cK^2 - 2c\vec{K} \cdot \vec{q}}{rt + cq^2} + \dots \right) \quad (73)$$

$$\approx \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \left[\frac{1}{(rt + cq^2)^2} + O(K^2/q^2) \right] \quad (74)$$

REMARK: It means we neglect terms of type $\phi^2 (\nabla \phi)^2$, $\phi^2 \nabla^2 \phi^2$, ...

Renormalized action up to second order

Introducing $\vec{k}' = \vec{k}\ell$ and $\hat{\phi}' = z^{-1}\hat{\phi}_<$,

$$\begin{aligned} S'[\hat{\phi}'] &= \frac{1}{2} \int^\Lambda \frac{d\vec{k}' \ell^{-d}}{(2\pi)^d} z^2 |\hat{\phi}'(\vec{k}')|^2 \left[rt_1 + c \left(1 - \frac{96}{c} \left(\frac{u}{4} \right)^2 \frac{d}{dk^2} A(\vec{k}; c, rt) \Big|_{k^2=0} \right) \frac{k'^2}{\ell^2} \right] \\ &\quad + \frac{u_1}{4} \int^\Lambda \frac{d\vec{k}'_1 d\vec{k}'_2 d\vec{k}'_3}{(2\pi)^{3d}} z^4 \ell^{-3d} \hat{\phi}'(\vec{k}'_1) \hat{\phi}'(\vec{k}'_2) \hat{\phi}'(\vec{k}'_3) \hat{\phi}'(-\vec{k}'_1 - \vec{k}'_2 - \vec{k}'_3) \end{aligned} \quad (75)$$

in which

$$rt_1 = rt + 12 \frac{u}{4} \int_{\Lambda/\ell}^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} + O(u^2) \quad (76)$$

$$u_1 = u - 36 \frac{u^2}{4} \int_{\Lambda/\ell}^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2)^2} \quad (77)$$

Renormalized action up to second order

Field renormalization can be fixed requiring canonical scaling

$$c' = \ell^{-d-2} z^2 c_1 = c, \text{ i.e.}$$

$$\ell^{-d-2} z^2 \left[1 - \frac{96}{c} \left(\frac{u}{4} \right)^2 \frac{d}{dk^2} A(\vec{k}; c, rt) \Big|_{k^2=0} \right] = 1 \quad (78)$$

which implies

$$z = \frac{\ell^{1+\frac{d}{2}}}{\left[1 - \frac{96}{c} \left(\frac{u}{4} \right)^2 \frac{d}{dk^2} A(\vec{k}; c, rt) \Big|_{k^2=0} \right]^{\frac{1}{2}}} \quad (79)$$

$$\approx \ell^{1+\frac{d}{2}} \left[1 + \frac{96}{2c} \left(\frac{u}{4} \right)^2 \frac{d}{dk^2} A(\vec{k}; c, rt) \Big|_{k^2=0} \right] \quad (80)$$

$$= \ell^{1+\frac{d}{2}} (1 + O(u^2)) \quad (81)$$

REMARK: Neglecting corrections means no anomalous dimension of the field, i.e. $\eta = 0$.

Renormalized action up to second order

For $\ell \approx 1$, momentum-shell integrals with spherical symmetry can be approximated as follows

$$\int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} f(k) = K_d \int_{\Lambda/\ell}^{\Lambda} q^{d-1} f(q) dq \approx K_d \left(1 - \frac{1}{\ell}\right) \Lambda^d f(\Lambda) \quad (82)$$

with $K_d = \frac{S_d}{(2\pi)^d} = \frac{2\pi^{d/2}}{(2\pi)^d (d/2-1)!}$. Hence,

$$rt_1 = rt + 12 \frac{u}{4} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} + O(u^2) \quad (83)$$

$$\approx rt + \frac{12K_d u}{4} \frac{\Lambda^d}{rt + c\Lambda^2} \left(1 - \frac{1}{\ell}\right) + O(u^2) \quad (84)$$

$$u_1 = u - 36 \frac{u^2}{4} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2)^2} \quad (85)$$

$$\approx u - 36 \frac{u^2}{4} \frac{K_d \Lambda^d}{(rt + c\Lambda^2)^2} \left(1 - \frac{1}{\ell}\right) \quad (86)$$

RG flow equations

Assuming $z = \ell^{1+d/2}$, the renormalized parameters read

$$rt'(\ell) = \ell^{-d} z^2 rt_1 \quad (87)$$

$$= \ell^2 \left[rt + \frac{u}{4} \frac{12K_d \Lambda^d}{rt + c\Lambda^2} \left(1 - \frac{1}{\ell}\right) \right] + O(u^2) \quad (88)$$

$$u'(\ell) = \ell^{-3d} z^4 u_1 \quad (89)$$

$$= \ell^{4-d} \left[u - \frac{36(u^2/4)K_d \Lambda^d}{(rt + c\Lambda^2)^2} \left(1 - \frac{1}{\ell}\right) \right] + O(u^3) \quad (90)$$

Setting $\ell = e^s$ and promoting parameters to be function of s , for an infinitesimal rescaling $e^{\delta s} \approx 1 + \delta s$, we have

$$\ell^a = 1 + a\delta s, \quad rt'(s) = rt(s) + r \frac{dt}{ds} \delta s, \quad u'(s) = u(s) + \frac{du}{ds} \delta s \quad (91)$$

RG flow equations

The expressions for the renormalized parameters become

$$rt + r \frac{dt}{ds} \delta s = (1 + 2\delta s) \left[rt + \frac{12K_d \Lambda^d}{rt + c\Lambda^2} \frac{u}{4} \delta s \right] + O(u^2) \quad (92)$$

$$u + \frac{du}{ds} \delta s = (1 + (4 - d)\delta s) \left[u - \frac{36K_d \Lambda^d}{(rt + c\Lambda^2)^2} \frac{u^2}{4} \delta s \right] + O(u^3) \quad (93)$$

Taking $\delta s \rightarrow 0$, we obtain a “dynamical system” which represents the *momentum-shell RG flow equations* (at $O(u^2)$)

$$\frac{dt}{ds} = 2t + \frac{12K_d \Lambda^d}{r(rt + c\Lambda^2)} \frac{u}{4} + O(u^2) \quad (94)$$

$$\frac{du}{ds} = (4 - d)u - \frac{36K_d \Lambda^d}{(rt + c\Lambda^2)^2} \frac{u^2}{4}. \quad (95)$$

RG flow equations

The RG flow equations admit two fixed points:

- the *Gaussian* fixed point at $(t_*, u_*) = (t_G, u_G) = (0, 0)$,
- a non-trivial fixed point at (t_*, u_*) with

$$\begin{aligned}rt_* &= -\frac{6K_d\Lambda^d}{4(rt_* + c\Lambda^2)}u_* \\ u_* &= \frac{(rt_* + c\Lambda^2)^2}{9K_d\Lambda^d}(4-d)\end{aligned}$$

We should linearize the system around the fixed points to investigate their stability properties.

REMARK: The non-trivial fixed point is meaningful as long as perturbative theory is valid, so we need u to be small.

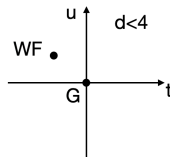
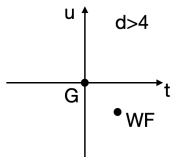
The ϵ -expansion and the Wilson-Fisher fixed point

Introduce $\epsilon = 4 - d$, and expand in powers of ϵ , so that we study the theory in the vicinity of d_{uc} .

The non-trivial fixed point becomes

$$\begin{aligned}u_* &= \frac{(rt_* + c\Lambda^2)^2}{9K_{4-\epsilon}\Lambda^{4-\epsilon}}\epsilon \approx \frac{c^2}{9K_4}\epsilon \\rt_* &= -\frac{6K_{4-\epsilon}\Lambda^{4-\epsilon}}{4(rt_* + c\Lambda^2)}u_* \approx -\frac{6K_4\Lambda^4}{4c\Lambda^2}\frac{c^2}{9K_4}\epsilon \approx -\frac{\Lambda^2 c}{6}\epsilon\end{aligned}\tag{96}$$

which is called **Wilson-Fisher** fixed point $(t_{\text{WF}}, u_{\text{WF}}) = \left(-\frac{\Lambda^2 c}{6r}\epsilon, \frac{c^2}{9K_4}\epsilon\right)$



The ϵ -expansion and the Wilson-Fisher fixed point

Linearizing the RG equations close to a fixed point we find

$$\frac{d}{ds} \begin{pmatrix} \delta t \\ \delta u \end{pmatrix} = J \begin{pmatrix} \delta t \\ \delta u \end{pmatrix} \quad (97)$$

with stability matrix

$$J = \begin{pmatrix} 2 - \frac{3u_* K_d \Lambda^d}{(rt_* + c\Lambda^2)^2} & \frac{3K_d \Lambda^d}{r(rt_* + c\Lambda^2)} \\ \frac{18ru_*^2 K_d \Lambda^d}{(t_* + c\Lambda^2)^3} & (4 - d) - \frac{18K_d \Lambda^d}{(rt_* + c\Lambda^2)^2} u_* \end{pmatrix} \quad (98)$$

The ϵ -expansion and the Wilson-Fisher fixed point

At the *Gaussian fixed point*,

$$J_G = \begin{pmatrix} 2 & \frac{3K_4\Lambda^2}{rc} \\ 0 & \epsilon \end{pmatrix} \quad (99)$$

in which we identify the eigenvalues $y_1 = 2$ and $y_2 = \epsilon$ with eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -\frac{3\Lambda^2}{4rc(2\pi)^2} \\ 1 \end{pmatrix} \quad (100)$$

At the *Wilson-Fisher fixed point* (in the $\epsilon \rightarrow 0$ limit),

$$J_{WF} \simeq \begin{pmatrix} 2 - \frac{3K_4\Lambda^4}{c^2\Lambda^4} \frac{c^2\epsilon}{9K_4} & \frac{3K_4\Lambda^4}{rc\Lambda^2} \\ \frac{18rK_4\Lambda^4}{c^3\Lambda^6} \frac{c^4\epsilon^2}{9^2K_4^2} & \epsilon - \frac{18K_4\Lambda^4}{c^2\Lambda^4} \frac{c^2\epsilon}{9K_4} \end{pmatrix} \quad (101)$$

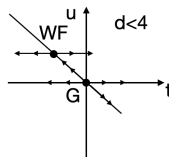
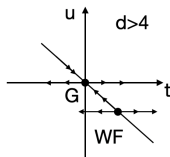
$$\approx \begin{pmatrix} 2 - \frac{\epsilon}{3} & \frac{3K_4\Lambda^2}{rc} \\ O(\epsilon^2) & -\epsilon \end{pmatrix}, \quad (102)$$

with $y_1 = 2 - \epsilon/3$, $y_2 = -\epsilon$ and same eigenvectors as before (at $O(1)$).

The ϵ -expansion and the Wilson-Fisher fixed point

In summary:

- if $d > 4$, the Gaussian fixed point is stable along direction v_2 while WF f.p. is unstable,
- if $d < 4$, the WF fixed point is stable along direction v_2 and the Gaussian one unstable.



The RG flow converges to the WF f.p. (non-MF critical behavior) instead of the Gaussian one (MF behavior) for $d < 4$.

The critical exponents at WF fixed point

At the WF fixed point we have $y_t = y_1 = 2 - \epsilon/3$, therefore

$$\nu = \frac{1}{y_t} = \frac{1}{2 - \epsilon/3} \approx \frac{1}{2} + \frac{\epsilon}{12} \quad (103)$$

then using scaling relations

$$\alpha = 2 - d\nu \approx 2 - \frac{4 - \epsilon}{2} \left(1 + \frac{\epsilon}{6}\right) \approx \frac{\epsilon}{6}. \quad (104)$$

From field renormalization $\eta = 0 + O(\epsilon^2)$, therefore

$$\gamma = (2 - \eta)\nu \approx 2\nu \approx 1 + \frac{\epsilon}{6}. \quad (105)$$

The critical exponents at WF fixed point

The rescaling of the external (uniform) field follows from

$$h\hat{\phi}(\vec{k}=0) = zh\hat{\phi}'(\vec{k}=0) \quad (106)$$

which implies $h' = zh = \ell^{1+\frac{d}{2}}h$ and finally,

$$y_h = 1 + \frac{d}{2} = 3 - \frac{\epsilon}{2}. \quad (107)$$

Then

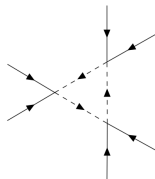
$$\beta = \frac{d - y_h}{y_t} = \frac{1 - \frac{\epsilon}{2}}{2 - \frac{\epsilon}{3}} \approx \frac{1}{2} \left(1 - \frac{\epsilon}{2}\right) \left(1 + \frac{\epsilon}{6}\right) = \frac{1}{2} - \frac{\epsilon}{6} \quad (108)$$

and

$$\delta = \frac{y_h}{d - y_h} = \frac{3 - \frac{\epsilon}{2}}{1 - \frac{\epsilon}{2}} \approx \left(3 - \frac{\epsilon}{2}\right) \left(1 + \frac{\epsilon}{2}\right) = 3 + \epsilon \quad (109)$$

Irrelevance of higher-order vertices

Suppose we consider diagrams generated of terms of the type $g_6 \phi(\vec{x})^6$, e.g.



$$g'_6 = \ell^{-5d} z^6 \tilde{g}_6 = \ell^{6-2d} \tilde{g}_6 = \ell^{6-2d} [g_6 - O(u^3, g_6^2)] \quad (110)$$

$$\implies \frac{dg_6}{ds} = (6 - 2d) g_6 + O(u^3, g_6^2) \quad (111)$$

$$= 2(-1 + \epsilon) g_6 + O(u^3, g_6^2). \quad (112)$$

that is the coupling is irrelevant under RG.

Field renormalization and exponent η at $O(\epsilon^2)$

Consider again the integral

$$\begin{aligned}
 \left. \frac{d}{dk^2} A(\vec{k}; c, rt) \right|_{k^2=0} &= \left. \frac{d}{dk^2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \frac{1}{(rt + cq_1^2)} \frac{1}{(rt + cq_2^2)} \frac{1}{rt + c(\vec{k} - \vec{q}_1 - \vec{q}_2)^2} \right|_{k^2=0} \\
 &= \left. \frac{d}{dk^2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_1}{(2\pi)^d} \frac{d\vec{q}_2}{(2\pi)^d} \frac{d\vec{q}_3}{(2\pi)^d} \frac{(2\pi)^d \delta(\vec{q}_3 - \vec{k} + \vec{q}_1 + \vec{q}_2)}{(rt + cq_1^2)(rt + cq_2^2)(rt + cq_3^2)} \right|_{k^2=0} \\
 &= \left. \frac{d}{dk^2} \int_{\Lambda/\ell}^{\Lambda} \prod_{i=1}^3 \frac{d\vec{q}_i}{(2\pi)^d} \frac{1}{(rt + cq_i^2)} \int d\vec{R} e^{i\vec{R} \cdot (\vec{q}_3 - \vec{k} + \vec{q}_1 + \vec{q}_2)} \right|_{k^2=0} \\
 &= \left. \frac{d}{dk^2} \int d\vec{R} \int_{\Lambda/\ell}^{\Lambda} \prod_{i=1}^3 \frac{d\vec{q}_i}{(2\pi)^d} \frac{e^{i\vec{R} \cdot \vec{q}_i}}{(rt + cq_i^2)} e^{-i\vec{R} \cdot \vec{k}} \right|_{k^2=0} \\
 &= \left. \frac{d}{dk^2} \int d\vec{R} \int_{\Lambda/\ell}^{\Lambda} \prod_{i=1}^3 \frac{d\vec{q}_i}{(2\pi)^d} \frac{e^{i\vec{R} \cdot \vec{q}_i}}{(rt + cq_i^2)} \left[1 - i\vec{R} \cdot \vec{k} - \frac{1}{2} (\vec{R} \cdot \vec{k})^2 + \dots \right] \right|_{k^2=0} \quad (113)
 \end{aligned}$$

we need approximations to avoid computing angular integrals.

Field renormalization and exponent η at $O(\epsilon^2)$

Using the approximation

$$(\vec{R} \cdot \vec{k})^2 = \left(\sum_{\alpha=1}^d R_{\alpha} k_{\alpha} \right)^2 = \sum_{\alpha, \beta=1}^d R_{\alpha} R_{\beta} k_{\alpha} k_{\beta} \quad (114)$$

$$\approx \sum_{\alpha} R_{\alpha}^2 k_{\alpha}^2 \quad (115)$$

$$\approx \frac{R^2}{d} \sum_{\alpha} k_{\alpha}^2 = \frac{R^2}{d} k^2 \quad (116)$$

we obtain

$$\left. \frac{d}{dk^2} A(\vec{k}; c, rt) \right|_{k^2=0} \approx -\frac{1}{2d} \int d\vec{R} R^2 \prod_{i=1}^3 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}_i}{(2\pi)^d} \frac{e^{i\vec{R} \cdot \vec{q}_i}}{(rt + cq_i^2)} \quad (117)$$

$$= -\frac{1}{2d} \int d\vec{R} R^2 \left[G_{0,>}(\vec{R}) \right]^3 \quad (118)$$

Field renormalization and exponent η at $O(\epsilon^2)$

For $d = 4$, we can use the explicit form

$$G_{0,>}(\vec{R}) = \frac{1}{(2\pi)^2} \frac{1}{cR^2} (J_0(R\Lambda/\ell) - J_0(R\Lambda)) \quad (119)$$

where the Bessel function $J_0(x)$ is an oscillating function decaying fast for $x > 1$.

Approximating $J_0(R\Lambda/\ell) - J_0(R\Lambda) \approx \theta(\ell/\Lambda - R) - \theta(1/\Lambda - R)$, then

$$\left. \frac{d}{dk^2} A(\vec{k}; c, rt) \right|_{k^2=0} \approx -\frac{S_4}{8(2\pi)^6 c^3} \int_{1/\Lambda}^{\ell/\Lambda} dR R^5 \frac{1}{R^6} \quad (120)$$

$$= -\frac{2\pi^2/\Gamma(2)}{8(2\pi)^6 c^3} \int_{1/\Lambda}^{\ell/\Lambda} \frac{dR}{R} \quad (121)$$

$$= -\frac{1}{4(8\pi^2)^2 c^3} \log \ell \quad (122)$$

Field renormalization and exponent η at $O(\epsilon^2)$

Inserting it into the expression for z , at the WF fixed point,

$$z = \ell^{1+\frac{d}{2}} \left[1 + \frac{96}{2c} \left(\frac{u_{WF}}{4} \right)^2 \frac{d}{dk^2} A(\vec{k}; c, rt) \Big|_{k^2=0} \right] \quad (123)$$

$$\approx \ell^{1+\frac{d}{2}} \left[1 - \frac{96}{2c} \left(\frac{\epsilon c^2}{4 \cdot 9K_4} \right)^2 \frac{1}{4(8\pi^2)^2 c^3} \log \ell \right] \quad (124)$$

$$\approx \ell^{1+\frac{d}{2}} \left[1 - \frac{\epsilon^2}{108} \log \ell \right] \quad (125)$$

$$\approx \ell^{1+\frac{d}{2} - \frac{\epsilon^2}{108}} \quad (126)$$

Then, comparing $G(r) \sim r^{2-d-\eta} \sim \ell^{2-d-\eta}$ with

$$G(r') = \zeta^{-2} G(r) = z^{-2} \ell^{2d} G(r) \implies G(r) \sim z^2 \ell^{-2d} \sim \ell^{2-d-\frac{\epsilon^2}{54}} \quad (127)$$

$$\implies \eta = \frac{\epsilon^2}{54}. \quad (128)$$

Renormalization of the $O(n)$ model

Consider the vector field $\vec{\phi} = \{\phi_a(\vec{x})\}_{a=1}^n$ with action

$$S[\vec{\phi}] = \int d\vec{x} \left[\frac{c}{2} (\nabla \vec{\phi})^2 + \frac{rt}{2} \vec{\phi} \cdot \vec{\phi} + \frac{u}{4} (\vec{\phi} \cdot \vec{\phi})^2 \right]. \quad (129)$$

$$= \int d\vec{x} \left[\frac{c}{2} \sum_{a=1}^n (\nabla \phi_a)^2 + \frac{rt}{2} \sum_{a=1}^n \phi_a^2 + \frac{u}{4} \left(\sum_a \phi_a^2 \right)^2 \right] \quad (130)$$

$$= \int d\vec{x} \left[\frac{c}{2} \sum_{a=1}^n \sum_{i=1}^d \left(\frac{\partial \phi_a}{\partial x_i} \right)^2 + \frac{rt}{2} \sum_{a=1}^n \phi_a^2 + \frac{u}{4} \left(\sum_a \phi_a^2 \right)^2 \right] \quad (131)$$

We focus on the continuous phase transition determining the onset of long-range correlations close to $d_{uc} = 4$.

Renormalization of the $O(n)$ model

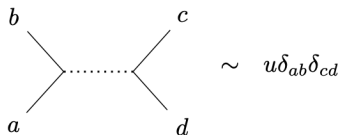
Let's treat perturbatively the interacting part

$$S_I = \frac{u}{4} \int d\vec{x} \left(\sum_a \phi_a^2 \right)^2 \quad (132)$$

$$= \frac{u}{4} \int d\vec{x} \left(\int \frac{d\vec{k}_1}{(2\pi)^d} \frac{d\vec{k}_2}{(2\pi)^d} \sum_a \hat{\phi}_a(\vec{k}_1) \hat{\phi}_a(\vec{k}_2) e^{i(\vec{k}_1 + \vec{k}_2) \cdot \vec{x}} \right)^2 \quad (133)$$

$$= \frac{u}{4} \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \sum_a \hat{\phi}_a(\vec{k}_1) \hat{\phi}_a(\vec{k}_2) \sum_b \hat{\phi}_b(\vec{k}_3) \hat{\phi}_b(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \quad (134)$$

which can be represented as the fourth-order vertex


$$\sim u \delta_{ab} \delta_{cd}$$

Renormalization of the $O(n)$ model

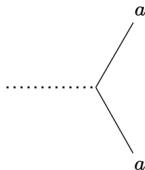
Using the Gaussian identity (*Hubbard-Stratonovich transformation*)

$$\int D\psi \, e^{-\frac{1}{2} \int d\vec{x} \psi(\vec{x})^2 + \int d\vec{x} \psi(\vec{x}) \Phi(\vec{x})} \propto e^{\frac{1}{2} \int d\vec{x} \Phi(\vec{x})^2} \quad (135)$$

with $\Phi(\vec{x}) = -i\sqrt{u/2} \sum_{a=1}^N \phi_a(\vec{x})^2$, we obtain

$$Z \propto \int D\phi D\psi \, e^{-\int d\vec{x} \left[\frac{\epsilon}{2} \sum_{a=1}^n (\nabla \phi_a)^2 + \frac{\epsilon}{2} \sum_{a=1}^n \phi_a^2 + i\sqrt{\frac{u}{2}} \psi(\vec{x}) \sum_{a=1}^N \phi_a(\vec{x})^2 + \frac{1}{2} \psi^2(\vec{x}) \right]} \quad (136)$$

with $\langle \psi(\vec{x}) \psi(\vec{x}') \rangle = \delta(\vec{x} - \vec{x}')$ and a vertex of type

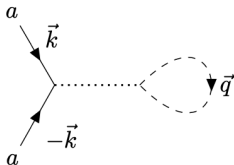


Renormalization of the $O(n)$ model

At order $O(u)$, we have two contributions to $\langle S_I \rangle$,

$$\begin{aligned}
 & 2 \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} \sum_a \hat{\phi}_{<,a}(\vec{k}_1) \hat{\phi}_{<,a}(\vec{k}_2) \\
 & \quad \times \left\langle \sum_b \hat{\phi}_{>,b}(\vec{k}_3) \hat{\phi}_{>,b}(\vec{k}_4) \right\rangle_{0,>} (2\pi)^d \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \\
 & = 2 \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \sum_a \hat{\phi}_{<,a}(\vec{k}) \hat{\phi}_{<,a}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{n}{rt+cq^2} \\
 & = 2n \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \left| \vec{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt+cq^2},
 \end{aligned} \tag{137}$$

which corresponds to

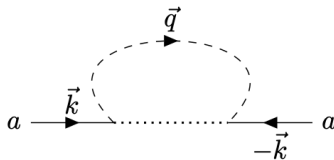


Renormalization of the $O(n)$ model

and

$$\begin{aligned}
 & 4 \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} \sum_a \sum_b \hat{\phi}_{<,a}(\vec{k}_1) \hat{\phi}_{<,b}(\vec{k}_2) \\
 & \quad \times \left\langle \hat{\phi}_{>,a}(\vec{k}_3) \hat{\phi}_{>,b}(\vec{k}_4) \right\rangle_{0,>} (2\pi)^d \delta \left(\sum_{i=1}^4 \vec{k}_i \right) \\
 & = 4 \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \sum_a \sum_b \hat{\phi}_{<,a}(\vec{k}) \hat{\phi}_{<,b}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{\delta_{ab}}{rt+cq^2}, \\
 & = 4 \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \left| \vec{\phi}_{<}(\vec{k}) \right|^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt+cq^2},
 \end{aligned} \tag{138}$$

which corresponds to

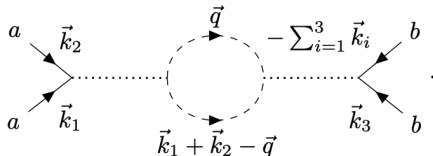


Renormalization of the $O(n)$ model

At order $O(u^2)$, three contributions to $\langle S_I^2 \rangle$:

$$\begin{aligned}
 & 8 \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} \int^\Lambda \frac{d\vec{k}_5 d\vec{k}_6 d\vec{k}_7 d\vec{k}_8}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_{i=1}^4 \vec{k}_i\right) (2\pi)^d \delta\left(\sum_{j=5}^8 \vec{k}_j\right) \\
 & \quad \times \sum_{a=1}^n \hat{\phi}_{<,a}(\vec{k}_1) \hat{\phi}_{<,a}(\vec{k}_2) \sum_{d=1}^n \hat{\phi}_{<,d}(\vec{k}_7) \hat{\phi}_{<,d}(\vec{k}_8) \\
 & \quad \times \sum_{b=1}^n \sum_{c=1}^n \left\langle \hat{\phi}_{>,b}(\vec{k}_3) \hat{\phi}_{>,c}(\vec{k}_5) \right\rangle_{0,>} \left\langle \hat{\phi}_{>,b}(\vec{k}_4) \hat{\phi}_{>,c}(\vec{k}_6) \right\rangle_{0,>} \\
 & = 8n \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \left[\vec{\phi}_{<}(\vec{k}_1) \cdot \vec{\phi}_{<}(\vec{k}_2) \right] \left[\vec{\phi}_{<}(\vec{k}_3) \cdot \vec{\phi}_{<}(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \right] \\
 & \quad \times \int^{\Lambda/\ell} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt+cq^2)(rt+c(\vec{k}_1+\vec{k}_2-\vec{q})^2)}
 \end{aligned} \tag{139}$$

which corresponds to

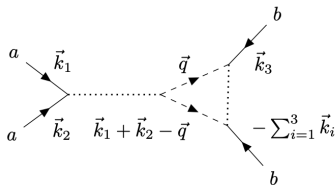


Renormalization of the $O(n)$ model

Then 2 symmetric contributions each one equal to

$$\begin{aligned}
 & 16 \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} \int^\Lambda \frac{d\vec{k}_5 d\vec{k}_6 d\vec{k}_7 d\vec{k}_8}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_{i=1}^4 \vec{k}_i\right) (2\pi)^d \delta\left(\sum_{j=5}^8 \vec{k}_j\right) \\
 & \times \sum_{a=1}^n \hat{\phi}_{<,a}(\vec{k}_1) \hat{\phi}_{<,a}(\vec{k}_2) \sum_{b=1}^n \sum_{c=1}^n \sum_{d=1}^n \hat{\phi}_{<,c}(\vec{k}_6) \hat{\phi}_{<,d}(\vec{k}_8) \\
 & \times \left\langle \hat{\phi}_{>,b}(\vec{k}_3) \hat{\phi}_{>,c}(\vec{k}_5) \right\rangle_{0,>} \left\langle \hat{\phi}_{>,b}(\vec{k}_4) \hat{\phi}_{>,d}(\vec{k}_7) \right\rangle_{0,>} \\
 & = 16 \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \left[\vec{\phi}_{<}(\vec{k}_1) \cdot \vec{\phi}_{<}(\vec{k}_2) \right] \left[\vec{\phi}_{<}(\vec{k}_3) \cdot \vec{\phi}_{<}(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \right] \\
 & \times \int^{\Lambda/\ell} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt+cq^2)(rt+c(\vec{k}_1+\vec{k}_2-\vec{q})^2)}
 \end{aligned} \tag{140}$$

that is

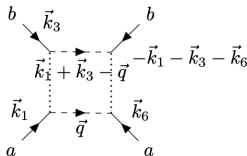


Renormalization of the $O(n)$ model

Finally, we have

$$\begin{aligned}
 & 32 \int^\Lambda \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_7 d\vec{k}_8}{(2\pi)^{4d}} \int^\Lambda \frac{d\vec{k}_3 d\vec{k}_4 d\vec{k}_5 d\vec{k}_6}{(2\pi)^{4d}} (2\pi)^d \delta\left(\sum_{i=1}^4 \vec{k}_i\right) (2\pi)^d \delta\left(\sum_{j=5}^8 \vec{k}_j\right) \\
 & \times \sum_{a=1}^n \sum_{b=1}^n \sum_{c=1}^n \sum_{d=1}^n \phi_{<,a}(\vec{k}_1) \phi_{<,b}(\vec{k}_3) \phi_{<,c}(\vec{k}_6) \phi_{<,d}(\vec{k}_8) \\
 & \times \left\langle \hat{\phi}_{>,a}(\vec{k}_2) \hat{\phi}_{>,c}(\vec{k}_5) \right\rangle_{0,>} \left\langle \hat{\phi}_{>,b}(\vec{k}_4) \hat{\phi}_{>,d}(\vec{k}_7) \right\rangle_{0,>} \\
 & = 32 \int^{\Lambda/\ell} \frac{d\vec{k}_1 d\vec{k}_3 d\vec{k}_6}{(2\pi)^{3d}} \left[\vec{\phi}_{<}(\vec{k}_1) \cdot \vec{\phi}_{<}(\vec{k}_6) \right] \left[\vec{\phi}_{<}(\vec{k}_3) \cdot \vec{\phi}_{<}(-\vec{k}_1 - \vec{k}_3 - \vec{k}_6) \right] \\
 & \times \int_{\Lambda/\ell}^\Lambda \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt+cq^2)(rt+c(\vec{k}_1+\vec{k}_2-\vec{q})^2)}
 \end{aligned} \tag{141}$$

which is



Renormalization of the $O(n)$ model

Putting all contributions together, we obtain an expression for the coarse-grained parameters,

$$rt_1 = rt + 4 \frac{u}{4} (n+2) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \quad (142)$$

$$u_1(\vec{k}_1 + \vec{k}_2) = u - \frac{4(n+8)u^2}{4} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2) \left(rt + c(\vec{k}_1 + \vec{k}_2 - \vec{q})^2 \right)} \quad (143)$$

$$\begin{aligned} &= u - (n+8)u^2 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(rt + cq^2)^2} \\ &\quad \times \left[1 - \frac{2\vec{q} \cdot (\vec{k}_1 + \vec{k}_2) - (\vec{k}_1 + \vec{k}_2)^2}{rt + cq^2} + \dots \right] \end{aligned} \quad (144)$$

Renormalization of the $O(n)$ model

Performing rescaling $\vec{k}' = \vec{k}\ell$ and renormalization $\phi'_a = z^{-1}\phi_a$ as done for the scalar theory, we arrive to the RG flow equations

$$r \frac{dt}{ds} = 2rt + \frac{4(n+2)K_d\Lambda^d}{rt+c\Lambda^2} \frac{u}{4} + O(u^2) \approx 2rt + \frac{n+2}{8\pi^2} \frac{\Lambda^{4-\epsilon}}{rt+c\Lambda^2} u + O(u^2) \quad (145)$$

$$\frac{du}{ds} = (4-d)u - \frac{4(n+8)K_d\Lambda^d}{(rt+c\Lambda^2)^2} \frac{u^2}{4} \approx \epsilon u - \frac{n+8}{8\pi^2} \frac{\Lambda^{4-\epsilon}}{(rt+c\Lambda^2)^2} u^2 \quad (146)$$

and finally, introducing $\tilde{u} = \Lambda^{-\epsilon} u$,

$$\begin{cases} r \frac{dt}{ds} = 2rt + \frac{n+2}{8\pi^2} \frac{\Lambda^4}{rt+c\Lambda^2} \tilde{u} \\ \frac{d\tilde{u}}{ds} = \epsilon \tilde{u} - \frac{n+8}{8\pi^2} \frac{\Lambda^4}{(rt+c\Lambda^2)^2} \tilde{u}^2. \end{cases} \quad (147)$$

Renormalization of the $O(n)$ model

The fixed point condition

$$\tilde{u}_* = \frac{8\pi^2 (rt_* + c\Lambda^2)^2}{(n+8)\Lambda^4} \epsilon \quad (148)$$

$$rt_* = -\frac{n+2}{8\pi^2} \frac{\Lambda^4}{rt_* + c\Lambda^2} \tilde{u}_*, \quad (149)$$

admits two solutions: 1) the Gaussian f.p. at $(t_G, \tilde{u}_G) = (0, 0)$ and 2) the *Heisenberg fixed point* $(t_H, \tilde{u}_H)$ at

$$\tilde{u}_H = \frac{8\pi^2 c^2}{(n+8)} \epsilon \quad (150)$$

$$rt_H = -\frac{n+2}{2(n+8)} c\Lambda^2 \epsilon. \quad (151)$$

Renormalization of the $O(n)$ model

The linearized dynamics in the proximity of a fixed point is governed by the properties of the Jacobian stability matrix, which at the Heisenberg f.p. is given by

$$J_H \approx \begin{pmatrix} 2 - \frac{n+2}{n+8}\epsilon & \frac{n+2}{8\pi^2 c} \Lambda^2 \\ O(\epsilon^2) & -\epsilon \end{pmatrix}, \quad (152)$$

Neglecting $O(\epsilon^2)$ corrections, The eigenvalues are

$$\begin{cases} y_1 = y_t = 2 - \frac{n+2}{n+8}\epsilon \\ y_2 = -\epsilon \end{cases} \quad (153)$$

Renormalization of the $O(n)$ model

The critical exponents can be computed from known scaling relations,

$$\nu = \frac{1}{y_1} = \left[2 - \frac{n+2}{n+8}\epsilon\right]^{-1} \simeq \frac{1}{2} + \frac{1}{4} \frac{n+2}{n+8}\epsilon \quad (154)$$

$$\alpha = 2 - d\nu = 2 - \frac{4-\epsilon}{2} \left(1 + \frac{1}{2} \frac{n+2}{n+8}\epsilon\right) \simeq \frac{4-n}{2(n+8)}\epsilon \quad (155)$$

$$\beta = \frac{d - y_h}{y_1} = \frac{d - \left(1 + \frac{d}{2}\right)}{y_1} \simeq \frac{1}{2} - \frac{3}{2(n+8)}\epsilon \quad (156)$$

$$\gamma = \frac{2y_h - d}{y_1} = 1 + \frac{n+2}{2(n+8)}\epsilon. \quad (157)$$

while the last exponent $\eta \approx \frac{n+2}{2(n+8)^2}\epsilon^2$.

Model with cubic anisotropy

Consider again a vector field $\vec{\phi} = \{\phi_a(\vec{x})\}_{a=1}^n$ with action

$$S[\vec{\phi}] = \int d\vec{x} \left[\frac{c}{2} (\nabla \vec{\phi})^2 + \frac{rt}{2} \vec{\phi}^2 + \frac{u}{4} \left(\sum_a \phi_a^2 \right)^2 + \frac{v}{4} \sum_a \phi_a^4 \right] \quad (158)$$

The term $\sum_a \phi_a^4$ is a *cubic anisotropy* causing rotational symmetry breaking and selection of some preferred axis.

We will:

- 1 determine the directions selected for $v \leq 0$;
- 2 find restrictions on u, v in order to have a finite order parameter;
- 3 determine the MF phase diagram in the (t, v) plane for $u > 0$;
- 4 apply momentum-shell RG to determine fixed points and their stability.

Model with cubic anisotropy: symmetry breaking

To find selected directions, we first consider $n = 2$ and set $|\vec{\phi}| = \bar{\phi}$.

When $v > 0$, the term $\sum_a \phi_a^4$ should be minimized,

$$\begin{cases} \min_{\phi_1, \phi_2} [\phi_1^4 + \phi_2^4] \\ \phi_1^2 + \phi_2^2 = \bar{\phi}^2 \end{cases} \quad (159)$$

which gives

$$\min_{\phi_1, \phi_2} [\phi_1^4 + \phi_2^4] = \min_{\phi_1, \phi_2} \left[(\phi_1^2 + \phi_2^2)^2 - 2\phi_1^2\phi_2^2 \right] \quad (160)$$

$$= \min_{\phi_1, \phi_2} [\bar{\phi}^4 - 2\phi_1^2\phi_2^2] \quad (161)$$

$$= \min_{\phi_1} [\bar{\phi}^4 - 2\phi_1^2(\bar{\phi}^2 - \phi_1^2)] \quad (162)$$

Model with cubic anisotropy: symmetry breaking

Setting $\alpha = \phi_1^2 / \bar{\phi}^2$,

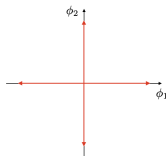
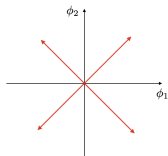
$$\min_{\alpha} \left[\bar{\phi}^4 - 2\alpha(1-\alpha)\bar{\phi}^4 \right] = \bar{\phi}^4 \min_{\alpha} [1 - 2\alpha(1-\alpha)] \quad (163)$$

which gives $-2 + 4\alpha = 0$, i.e. $\alpha = 1/2$.

Field directions selected by symmetry breaking are $\vec{\phi} = \left(\pm \frac{\bar{\phi}}{\sqrt{2}}, \pm \frac{\bar{\phi}}{\sqrt{2}} \right)$.

For $v < 0$, the same analysis gives $\max_{\alpha} [1 - 2\alpha(1-\alpha)]$, i.e. $\alpha = 0, 1$.

The selected directions are given by $\vec{\phi} = \left(\pm \bar{\phi}, 0 \right), \left(0, \pm \bar{\phi} \right)$.



Model with cubic anisotropy: MF phase diagram

We assume $u > 0$ and analyze the phase diagram in the (t, v) plane. The equilibrium state is uniform, where $\bar{\phi}$ solves saddle-point equations

$$0 = \begin{cases} rt\bar{\phi} + u\bar{\phi}^3 + \frac{v}{n}\bar{\phi}^3 & v > 0 \\ rt\bar{\phi} + u\bar{\phi}^3 + v\bar{\phi}^3 & v < 0 \end{cases}$$

which give $\bar{\phi} = 0$ or

$$\Rightarrow \bar{\phi}^2 = \begin{cases} -\frac{rt}{(u + \frac{v}{n})} & v > 0 \\ -\frac{rt}{(u + v)} & v < 0 \end{cases}$$

Since

$$S_{\text{sp}}[\bar{\phi}] = \begin{cases} \frac{rt}{2}\bar{\phi}^2 + \frac{u}{4}\bar{\phi}^4 + \frac{v}{4n}\bar{\phi}^4 & v > 0 \\ \frac{rt}{2}\bar{\phi}^2 + \frac{u}{4}\bar{\phi}^4 + \frac{v}{4}\bar{\phi}^4 & v < 0 \end{cases}$$

solutions $\bar{\phi} \neq 0$ identify global minima if $u + v/n > 0$ (for $v > 0$) or $u + v > 0$ (for $v < 0$).

Model with cubic anisotropy: MF phase diagram

For $v > 0$, continuous transition from *diagonal order* to disorder at $t = 0$.

For $v < 0$ and $v > -u$, the ordered phase for $t < 0$ has *cubic-axis order*.

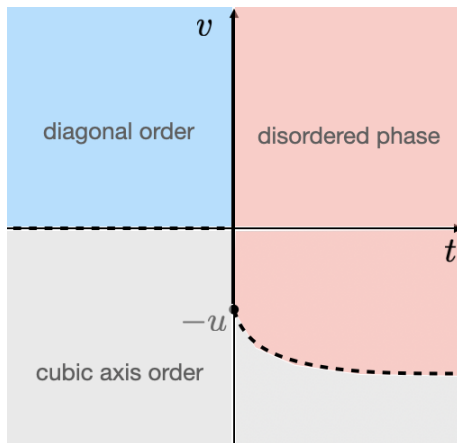
When $t > 0$ and $v + u < 0$, a line of first-order phase transitions appears. Imposing $S_{\min}[\bar{\phi}] = S_{\min}[0] = 0$, we find (introducing a regularizing term $\frac{u_6}{6}(\bar{\phi}^2)^3 > 0$)

$$\begin{cases} rt + (u + v) \bar{\phi}^2 + u_6 \bar{\phi}^4 = 0 \\ rt + \frac{u+v}{2} \bar{\phi}^2 + \frac{u_6}{3} \bar{\phi}^4 = 0 \end{cases} \implies \bar{\phi}^2 = -\frac{3}{4} \frac{u + v}{u_6}$$

Moreover,

$$0 = rt - \frac{3(u + v)^2}{4u_6} + \frac{9}{16} u_6 \left(\frac{u + v}{u_6} \right)^2 \implies rt_* = \frac{3(u + v)^2}{16u_6}.$$

Model with cubic anisotropy: MF phase diagram



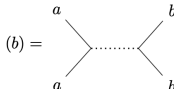
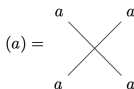
We focus on the second order phase transition at $t = 0$.

Model with cubic anisotropy: perturbative expansion

The action in the momentum space is our starting point,

$$\begin{aligned} S[\hat{\phi}] = & \int \frac{d\vec{k}}{(2\pi)^d} \left(\frac{rt + ck^2}{2} \right) \vec{\phi}(\vec{k}) \cdot \vec{\phi}(-\vec{k}) \\ & + \frac{u}{4} \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \sum_{a,b=1}^n \phi_a(\vec{k}_1) \phi_a(\vec{k}_2) \phi_b(\vec{k}_3) \phi_b(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \\ & + \frac{v}{4} \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3}{(2\pi)^{3d}} \sum_{a=1}^n \phi_a(\vec{k}_1) \phi_a(\vec{k}_2) \phi_a(\vec{k}_3) \phi_a(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \end{aligned}$$

The non-gaussian terms generate the two vertices, which both contribute to renormalize parameters rt as well as u and v :



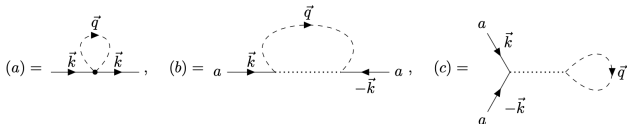
Model with cubic anisotropy: $O(u, v)$ corrections

Three corrections to the propagator at first order in u and v :

$$(a) = 6 \left(\frac{v}{4} \right) \sum_a \hat{\phi}_{<,a}(\vec{k}) \hat{\phi}_{<,a}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \quad (164)$$

$$(b) = 4 \left(\frac{u}{4} \right) \sum_a \hat{\phi}_{<,a}(\vec{k}) \hat{\phi}_{<,a}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \quad (165)$$

$$(c) = 2n \left(\frac{u}{4} \right) \sum_a \hat{\phi}_{<,a}(\vec{k}) \hat{\phi}_{<,a}(-\vec{k}) \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \quad (166)$$



Model with cubic anisotropy: critical temperature shift

Computing the shell integrals, we obtain

$$rt_1 = rt + \frac{K_d \Lambda^d}{(rt + c\Lambda^2)} [3v + (n+2)u] \delta s \quad (167)$$

which leads to the RG flow equation

$$r \frac{dt}{ds} = 2rt + \frac{K_d \Lambda^d}{(rt + c\Lambda^2)} [3v + (n+2)u] + O(u^2, v^2, uv) \quad (168)$$

The fixed point for t

$$rt_* = -\frac{K_d \Lambda^d}{2(rt + c\Lambda^2)} [3v_* + (n+2)u_*] \quad (169)$$

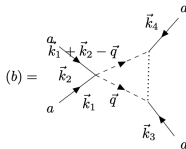
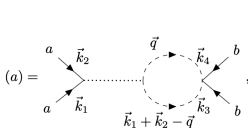
depends on those for u and v .

Model with cubic anisotropy: $O(u^2, v^2, uv)$ corrections

Corrections at $O(v^2)$ and $O(u^2)$ are those we already know (1 + 3 diagrams). Instead at $O(uv)$ we have

$$\begin{aligned}
 (a) = & 24 \times 2 \frac{uv}{4^2} \sum_a \hat{\phi}_{<,a}(\vec{k}_1) \hat{\phi}_{<,a}(\vec{k}_2) \sum_b \hat{\phi}_{<,b}(\vec{k}_3) \hat{\phi}_{<,b}(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \\
 & \times \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \frac{1}{rt + c(\vec{k}_1 + \vec{k}_2 - \vec{q})^2}
 \end{aligned} \tag{170}$$

$$\begin{aligned}
 (b) = & 48 \times 2 \frac{uv}{4^2} \sum_a \hat{\phi}_{<,a}(\vec{k}_1) \hat{\phi}_{<,a}(\vec{k}_2) \hat{\phi}_{<,a}(\vec{k}_3) \hat{\phi}_{<,a}(-\vec{k}_1 - \vec{k}_2 - \vec{k}_3) \\
 & \times \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \frac{1}{rt + c(\vec{k}_1 + \vec{k}_2 - \vec{q})^2}
 \end{aligned} \tag{171}$$



Model with cubic anisotropy: $O(u^2, v^2, uv)$ corrections

The shell integral is approximated by neglecting momentum dependence,

$$\begin{aligned} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{rt + cq^2} \frac{1}{rt + c(\vec{k}_1 + \vec{k}_2 - \vec{q})^2} &\approx \frac{K_d \Lambda^d}{(rt + c\Lambda^2)^2} \left(1 - \frac{1}{\ell}\right) \\ &= C_d \delta s \end{aligned} \quad (172) \quad (173)$$

We find the coarse-grained parameters

$$u_1 = u - \left[(n+8)u^2 + 6uv\right] C_d \delta s \quad (174)$$

$$v_1 = v - \left[9v^2 + 12uv\right] C_d \delta s \quad (175)$$

Model with cubic anisotropy: RG flow equations

Performing momentum rescaling and field renormalization, with $z = \ell^{1+d/2}$,

$$u' = \ell^{-3d} z^4 u_1 = e^{(-3d+4+2d)\delta s} \left[u - \left[(n+8)u^2 + 6uv \right] C_d \delta s \right] \quad (176)$$

$$= (1 + (4-d)\delta s) \left[u - \left[(n+8)u^2 + 6uv \right] C_d \delta s \right] \quad (177)$$

$$v' = \ell^{-3d} z^4 v_1 = e^{(-3d+4+2d)\delta s} \left[v - \left[9v^2 + 12uv \right] C_d \delta s \right] \quad (178)$$

$$= (1 + (4-d)\delta s) \left(v - \left[9v^2 + 12uv \right] C_d \delta s \right) \quad (179)$$

and finally we obtain the RG flow equations (with $\epsilon = 4 - d$ and $C_4 \approx K_4/c^2$),

$$\begin{cases} \frac{du}{ds} = \epsilon u - C_4 \left[(n+8)u^2 + 6uv \right] \\ \frac{dv}{ds} = \epsilon v - C_4 \left[9v^2 + 12uv \right] \end{cases} \quad (180)$$

Model with cubic anisotropy: fixed points

The fixed point condition reads

$$\epsilon u = C_4 \left[(n+8)u^2 + 6uv \right] \quad (181)$$

$$\epsilon v = C_4 \left[9v^2 + 12uv \right] \quad (182)$$

One possible solution is the usual *Gaussian fixed point* $u_G = v_G = 0$ (and $t_G = 0$).

Defining $x = u/v$, it implies

$$x = \frac{(n+8)u^2 + 6uv}{9v^2 + 12uv} \quad (183)$$

$$= \frac{(n+8)x^2 + 6x}{9 + 12x} \quad (184)$$

which admits solutions $x = 0$, $x = +\infty$ and $x = 3/(n-4)$.

Model with cubic anisotropy: fixed points

- 1 The case $x = +\infty$ corresponds to the *Heisenberg fixed point* $v_H = 0$, $u_H = \frac{\epsilon c^2}{(n+8)K_4}$.
- 2 The case $x = 0$ corresponds to the *Ising fixed point* $u_I = 0$ and $v_I = \frac{\epsilon c^2}{9K_4}$.
- 3 In the case $x = 3/(n-4)$, we find

$$\begin{aligned}\epsilon u &= C_4 \left[(n+8)u^2 + 6uv \right] \\ &= C_4 \left[(n+8) + 2(n-4) \right] u^2 \\ \implies u &= \frac{\epsilon}{3nC_4} = \frac{\epsilon c^2}{3nK_4}\end{aligned}$$

and $v = (n-4)/3u = \frac{(n-4)c^2}{9nK_4}\epsilon$. This is known as the *Cubic fixed point*, $(u_C, v_C) = \left(\frac{\epsilon c^2}{3nK_4}, \frac{(n-4)c^2}{9nK_4}\epsilon \right)$.

Model with cubic anisotropy: stability analysis

Linearizing the flow equations around the fixed points, we obtain the stability matrix

$$J = \begin{pmatrix} \epsilon - C_4 [2(n+8)u_* + 6v_*] & -6C_4 u_* \\ -12C_4 v_* & \epsilon - C_4 [18v_* + 12u_*] \end{pmatrix} \quad (185)$$

- At the Gaussian fixed point, we have

$$J_G = \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon \end{pmatrix} \quad (186)$$

with $y_1 = y_2 = \epsilon$ along the directions associated with u and v .

- At the Ising fixed point,

$$J = \begin{pmatrix} \frac{\epsilon}{3} & 0 \\ -\frac{4\epsilon}{3} & -\epsilon \end{pmatrix} \quad (187)$$

with eigenvalues $y_1 = \epsilon/3$ and $y_2 = -\epsilon$.

Model with cubic anisotropy: stability analysis

- At the Heisenberg fixed point,

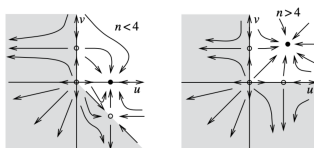
$$J = \begin{pmatrix} -\epsilon & -\frac{6\epsilon}{(n+8)} \\ 0 & \frac{n-4}{n+8}\epsilon \end{pmatrix}$$

with eigenvalues $y_1 = -\epsilon$ and $y_2 = \frac{n-4}{n+8}\epsilon$. Stability changes for $n \gtrless 4$.

- At the Cubic fixed point,

$$J = \begin{pmatrix} -\frac{n+8}{3} & -2 \\ -4\frac{(n-4)}{3} & 4-n \end{pmatrix} \frac{\epsilon}{n}$$

with $y_1 = \frac{4-n}{3n}\epsilon$ and $y_2 = -\epsilon$. Both directions are stable for $n > 4$, while one direction is unstable for $n < 4$.



from: M. Kardar, Stat. Phys. Fields